Kushal Agrawal

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Education

University of Wisconsin-Madison

Bachelor of Science in Computer Engineering, Computer Science ***(GPA: 3.77/4.00)*** *May 2026*

Coursework: Operating Systems, Data Structures & Algorithms, Object-Oriented Programming, Microprocessor Systems, Distributed Systems, Computer Architecture, High Performance Computing, Human-Computer Interaction, Probability

Experience

Endress+Hauser Group Greenwood, IN

Software Engineer Intern *May 2025 – Present*

- Engineered a Dockerized full-stack Netilion IIoT simulator (React/TypeScript, Python FastAPI) with **30+ REST endpoints, 5 simulation patterns** and **200+ health codes**, streaming real-time data for hardware-free demos.
- Built a wireless test platform in C and Python using an nRF54L15 BLE 6.2 dev kit and ESP32-C6 gateway, processing telemetry and generating latency and Channel Sounding datasets for SGC200.
- Conducted performance testing and QA for the Digital Commissioning App; documented workflows and proposed UI/UX and offline-first architecture changes tied to **$180K–$400K** global initiatives.
- Designed OPC UA and REST-based data pipes bridging Netilion IIoT with Snowflake via edge middleware clients.

Software Engineer Intern *May 2024 – August 2024*

- Built algorithms to analyze **1,100+** Salesforce dashboards for duplicates, then deployed a TypeScript/React naming web tool—integrated it as an extension—and proposed it across **10** departments, potentially saving **$40K** annually.
- Engineered an IoT pipeline by reading a 4–20 mA industrial temperature probe with a microcontroller, converting current-to-temperature in JavaScript/MakeCode, and streaming the data to ThingSpeak for real-time visualization.

UW-Madison Department of Electrical and Computer Engineering Madison, WI

Undergraduate Teaching Assistant *September 2023 – May 2025*

- Led office hours for **740** students in Machine Organization (ECE 354) and Intro to Computer Engineering (ECE 252)
- Instructed ECE 354 concepts: C programming, assembly, memory hierarchy, compilation, and signaling.
- Instructed ECE 252 concepts: Digital Logic Design, Processor Models, RTL design and Machine Language.

Larsen & Toubro Defence IC Mumbai, India

Software Engineer Intern *May 2023 – August 2023*

- Implemented data visualization dashboards using Power BI and MySQL to analyze **100,000 employee task tickets** from **5 business units** over **2 years**, providing actionable insights into task completion and performance trends.
- Developed and deployed a robust hit counter for 4 internal intranet portals using C#, HTML, and the .NET framework, tracking usage patterns of **4,000+ employees** to drive data-informed website optimizations.

Projects and Leadership

FileHawk.net - Local Semantic Search | *Electron, Python, ChromaDB, SentenceTransformers* | filehawk.net

- Architected a cross-platform desktop app (Electron + React + Flask) that searches **50GB+** local data with sub-second responses; two-stage retrieval (ChromaDB vectors + rerank) reduced file searching time by **75%**.
- Delivered high-precision semantic search using dual SentenceTransformers (MS MARCO / All-MiniLM) and a proprietary Gist ranker; intelligent chunking + caching drove **95%** relevance across **25+ file types.**
- Built real-time, incremental indexing (Watchdog + SHA-256 change detection) handling **10k+ file changes/hr**.

High-Frequency Currency Arbitrage Engine | *C++, OpenMP, CUDA, Python, Real-Time Data, CMake* | GitHub

- Built a parallel FX arbitrage engine analyzing **86,400 per-second timestamps** across **19 currency pairs**, modeling markets as dynamic graphs for real-time detection.
- Detected **680 arbitrage opportunities** using GPU acceleration, achieving **0.71% daily ROI** in simulation with position sizing and portfolio tracking.
- Leveraged parallelization to cut detection latency by **3×** and achieve a **3.5× speedup** in arbitrage cycle processing.

Executive Board Member | *Engineering Expo* September 2022 – May 2025

- Organized a large-scale STEM outreach event at UW-Madison, attracting over **5,000 visitors annually**.
- Managed logistics for **1,200 students** from 20 schools, coordinating schedules, transportation, and communication.

Technical Skills

Languages: C, C++, Python, Java, C#, TypeScript, JavaScript, Bash, SQL, HTML5/CSS3, CUDA, Assembly, LaTeX
Web & App Development: React, Electron, Next.js, Node.js, FastAPI, Flask, REST APIs, TailwindCSS, Postman
Machine Learning & AI: PyTorch, TensorFlow, SentenceTransformers, HuggingFace, ChromaDB, NumPy, Pandas
Tools: Git, GitHub, AWS, GCP, Docker, Snowflake, CMake, WebSockets, Slurm, OpenMP, Linux, x86, xv6, FUSE